F26 Stat 134 Practice 1 Midterm *Solution*

Print your name: _____

SID Number: _____

Exam Information and Instructions:

- You will have 110 minutes to take this exam.
- The exam will be scored out of 50 points.
- Closed book/notes, calculator, or collaborating.
- To receive full credit, you must show your work or brief reasoning in every answer.
- Unless stated otherwise, **you may leave answers as unsimplified numerical and algebraic expressions**, and in terms of the Normal c.d.f. Φ. Finite sums are fine, but simplify any infinite sums.
- We will be using Gradescope to grade this exam. Write any work you want graded on the front of each page, in the space below each question.
- Do **NOT** write your final solution on a scratch paper. It won't be graded. If additional space is required, please utilize the blank page located at the end of this exam paper. When doing so, please write the phrase "Solution continues at page X" at the bottom of the page. Additionally, please indicate on the blank page which question you are continuing your solution for.

I certify that all materials in the enclosed exam are my own original work.

Sign your name: _____

1. [10 pts]**(Markov, Chebychev, variance-expectation formula, inclusion of probability)** You are given a random variable X such that $\mathbb{E}[X] = 3$, $\mathbb{E}[X^2] = 10$. Recall that Chebychev's inequality states that

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq k \cdot sd(X)) \leq \frac{1}{k^2}.$$

   (a) Use Chebychev's inequality to get an upper bound of the tail probability $\mathbb{P}(X \geq 5)$.

   (b) You are given another piece of information that $\mathbb{P}(X \leq 0) = 0.05$. Use this additional information to get a tighter, i.e. smaller, upper bound of $\mathbb{P}(X \geq 5)$.

Solution

(a)

$$\mathbb{P}(X \geq 5) \leq \boxed{\frac{1}{4}}$$

(b)

$$\mathbb{P}(X \geq 5) \leq \boxed{0.2}$$

2. [10 pts]**(Hypergeometric, multiplication rule, Expectation with Indicators)** Ten children stand in a line to grab goodies out of a bag. The bag has 100 total goodies, 40 of which are candy and 60 of which are gum. While remaining in line, the children each grab five goodies at random from the bag *without replacement*. Let W be the number of adjacent pairs of children who *both* get exactly two pieces of candy. For example, if the 1st, 3rd, 4th and 5th children each get exactly two pieces of candy while the rest of the children get some other number of pieces of candy, then $W = 2$. Find $\mathbb{E}[W]$.

Solution

$$\mathbb{E}[W] = \boxed{9 \cdot \frac{\binom{40}{2}\binom{60}{3}}{\binom{100}{5}} \cdot \frac{\binom{38}{2}\binom{57}{3}}{\binom{95}{5}}}$$

3. [10 pts]**(inclusion exclusion, independence, binomial distribution)** Suppose in a school with 100 students, there are 60 students using Facebook, and 80 students using either Facebook or Twitter. If we choose a student at random, the event that the student uses Facebook (denote as F) and the event of using Twitter (denote as T) are independent.

   (a) What is the probability that a randomly-chosen student uses Twitter?

   (b) Suppose we pick 20 students **with** replacement. What's the chance that exactly 15 of them use Facebook *or* Twitter?

   Solution

   (a)

$$P(T) = 0.5$$

   (b)

$$\boxed{\mathbb{P}(X = 15) = \binom{20}{15} \left(\frac{80}{100}\right)^{15} \left(\frac{20}{100}\right)^{5}}$$

4. [10 pts]**(Theorem of Total Probability)**
Let U_1, U_2, U_3,...., U_n be n urns such that urn U_k contains k white and $k^2 - k$ black balls, $k = 1, ..., n$. Consider the random experiment of selecting an urn and drawing a ball out of it at random. Suppose the probability of selecting urn U_k is proportional to k (i.e $\mathbb{P}(U_k) = ck$ for some constant c).

a) Find the constant c as a function of n.

b) Find the probability that the ball drawn is black.

Solution

a)

$$c = \frac{2}{n(n+1)}$$

b)

$$\mathbb{P}(B) = \boxed{\frac{n-1}{n+1}}$$

5. [10 pts]**(CDF, exponential, uniform)** Beloved Stat 134 students Jonny and Ethan have very different study habits for the final. Jonny's study time is exponentialy distributed with rate $\lambda = \frac{1}{2}$ (i.e. average study time is 2 hours). Ethan's study time is uniformly distributed between 0 and 4 hours (also average 2 hours). You may assume Jonny and Ethan study independently. What is the chance the first quitter stops studying between 2 and 3 hours.

Solution
The answer is $F(3) - F(2) = e^{-1}/2 - e^{-3/2}/4$.

Blank page. Please indicate which question you are answering for.

Discrete

name and range	$P(k) = P(X = k)$ for $k \in$ range	mean	variance
uniform on $\{a, a+1, ..., b\}$	$\frac{1}{b-a+1}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$
Bernoulli(p) on $\{0, 1\}$	$P(1) = p$; $P(0) = 1-p$	p	$p(1-p)$
binomial(n,p) on $\{0, 1, ..., n\}$	$\binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
Poisson(μ) on $\{0, 1, 2, ...\}$	$\frac{e^{-\mu}\mu^k}{k!}$	μ	μ
hypergeometric(n,N,G) on $\{0, ..., n\}$	$\frac{\binom{G}{k}\binom{N-G}{n-k}}{\binom{N}{n}}$	$\frac{nG}{N}$	$n\left(\frac{G}{N}\right)\left(\frac{N-G}{N}\right)\left(\frac{N-n}{N-1}\right)$
geometric(p) on $\{1, 2, 3, ...\}$	$(1-p)^{k-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
geometric(p) on $\{0, 1, 2, ...\}$	$(1-p)^k p$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$
negative binomial(r,p) on $\{0, 1, 2, ...\}$	$\binom{k+r-1}{r-1} p^r (1-p)^k$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$

Continuous

† undefined

name	range	density $f(x)$ for $x \in$ range	c.d.f $F(x)$ for $x \in$ range	Mean	Variance
uniform(a,b)	(a, b)	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
normal(0,1)	$(-\infty, \infty)$	$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$	$\Phi(x)$	0	1
normal(μ,σ^2)	$(-\infty, \infty)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(x-\mu)^2/\sigma^2}$	$\Phi(\frac{x-\mu}{\sigma})$	μ	σ^2
exponential(λ) = gamma(1,λ)	$(0, \infty)$	$\lambda e^{-\lambda x}$	$1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
gamma(r,λ)	$(0, \infty)$	$\Gamma(r)^{-1} \lambda^r x^{r-1} e^{-\lambda x}$	$1 - e^{-\lambda x} \sum_{k=0}^{r-1} \frac{(\lambda x)^k}{k!}$ for integer r	$\frac{r}{\lambda}$	$\frac{r}{\lambda^2}$
chi-square(n) = gamma($\frac{n}{2}$,$\frac{1}{2}$)	$(0, \infty)$	$\Gamma(\frac{n}{2})^{-1} (\frac{1}{2})^{\frac{n}{2}} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}$	as above for $\lambda = \frac{1}{2}$, $r = \frac{n}{2}$ if n is even	n	$2n$
Rayleigh	$(0, \infty)$	$x e^{-\frac{1}{2}x^2}$	$1 - e^{-\frac{1}{2}x^2}$	$\sqrt{\frac{\pi}{2}}$	$\frac{4-\pi}{2}$
beta(r,s)	$(0, 1)$	$\frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)} x^{r-1}(1-x)^{s-1}$	see Exercise 4.6.5 for integer r and s	$\frac{r}{r+s}$	$\frac{rs}{(r+s)^2(r+s+1)}$
Cauchy	$(-\infty, \infty)$	$\frac{1}{\pi(1+x^2)}$	$\frac{1}{2} + \frac{1}{\pi} \arctan(x)$	†	†
arcsin = beta(1/2,1/2)	$(0, 1)$	$\frac{1}{\pi\sqrt{x(1-x)}}$	$\frac{2}{\pi} \arcsin(\sqrt{x})$	$\frac{1}{2}$	$\frac{1}{8}$